

Towards a Smart Interface-based Automated Learning Environment Through Social Media for Disaster Management and Smart Disaster Education

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Abstract. This article discusses the problem of integrating artificial intelligence technologies, social networks, Disaster Management, neural learning and Smart Education Disaster. It consists of an automated learning environment that is based on an extension of the Real-Time Alert Model used for natural and anthropogenic disaster management that is integrating encapsulations from multiple sources and retrieve information by combining multiple search results. This experience forms a background for our automated learning environment and provides some good ideas for that.

Finally, we summarize the main features of this approach, provide some references to related work, and discuss some issues with artificial intelligence, social media and disaster management.

Keywords: Automated learning, Data mining, Sequential pattern mining, Deep learning, Relevant content, Alert, Streaming, Disaster management

1 Introduction

In recent years, the frequency of large-scale disasters has increased worldwide. The Boumerdes earthquake [?] of May 21, 2003 and the Chlef earthquake (Algeria), for example, which occurred in 1980 (7.3 degrees on the Richter scale) caused 2,633 deaths and caused the collapse of 20,000 homes. Every time a disaster occurs, many recognize the importance of disaster management. The majority of disasters have huge and lasting impacts on human lives and societies. In order to minimize the consequences of such events, it is essential to prepare

and develop resilient communities by educating them with the appropriate and up-to-date information needed to cope with disasters. We will examine here how individuals, organizations and communities develop their skills in response to unforeseen events, whether in preparation, in disaster management or in the institutional aspect.

Our analytical framework is guided by the important social phenomenon that occurs in almost every disaster and the vastness of activity that brings this observation of evidence to wonder. Disasters such as earthquakes, floods, explosions, terrorist acts and tsunamis result in catastrophic human suffering, loss of property and other negative consequences [?].

People use social networking to post situational updates in various forms such as text messages, images and videos. Numerous studies [?] have shown that this online information is useful for a quick response to disaster management. These analyzes of the use of social networking in disasters have identified a distinct role for local users of the event (or "in the field"), who are more likely to generate useful information to improve situational awareness [?].

Thanks to the improvement of storage technics (big data) and the ease of use, the participatory culture and the interoperability of end-users (Web 2.0), we are being confronted with information overload stored, notably the Web which is a popular and interactive way of disseminating information today. It is notably immense, diverse and dynamic. This new era is defined by the rapid development of knowledge and that traditional learning initiatives have not been able to cope with the growing complexity and rapid changes of the new knowledge society. Therefore, a new vision of learning is needed, namely Automated Learning. Its aim is to discover, in large amounts of data, valuable information that can help understand these data or predict the behavior of future data. This process involves examining large amounts of data (stored in warehouses or in streaming) using statistical, mathematical or eventually AI techniques. This knowledge, unknown at first, may be correlations, patterns or general trends in that data. Automated Learning is based on the idea of analyzing problems to understand their principles and develop appropriate mathematical models. For that, experimental data are required to verify the correction of the system or the estimation of some difficult parameters to mathematical modeling [?]. A fundamental shift towards a more open and learner-centered learning model is needed where Automated Learning uses various manual and automatic tools: we start by describing the data, summarize their statistical attributes (means, variances, covariance), visualize them using the curves, the graphs, the diagrams, and finally look for the significant links potential between variables (such as values that are repeated). But the description of the data alone does not provide a plan of action. We must build a prediction model based on the information discovered, and we validate this model on other data. Over the past decade, it has been argued that Automated Learning could meet the needs of the new knowledge society and transform learning. However, in the majority of cases, systems do not have principles understood or are too complex for mathematical modeling. Automated Learning, which suggests complex and unimaginable specific

impacts on the society and more broadly the economy, remains primarily in the field of computer science, but is closely linked to cognitive science, neuroscience, biology and psychology and could lead to the crossroads of systems, nanotechnology, biotechnology and cognitive science, leading to artificial intelligence with a wider base.

Smart Disaster education plays a key role in encouraging community members to develop disaster management skills. In addition, these members also share their knowledge and help one another in disaster preparedness, like with other disaster education techniques, namely, Evacuation drills, Games Based-Evacuation Drill [?], Paradigmatic Tourism [?], Dark Tourism, Penumbral Tourism, Mixed Realistic Games or Alternative Realistic Games (ARG) [?] and tower defense game [?], or other evacuation techniques as Simulations.

In this article, we present the theoretical, design, implementation and evaluation details of an environment, a learning framework whose main objective is to help users become familiar with all of these different learning models using a wide variety of media and digital data available in this environment. This automated learning environment is fundamentally personal, social, distributed, ubiquitous, flexible and dynamic.

2 Background and Related Works

This section consists of studying Automated Learning, Disaster and Disaster Management, Social Networking and Disaster Education.

2.1 Automated Learning

Learning is a set of mechanisms leading to the acquisition of know-how and knowledge. But, Automated learning is a branch of Artificial Intelligence (AI) that deals with the development of algorithms that make capable of accomplishing complex tasks without having been explicitly programmed for that purpose, making extensive use tools and concepts of AI, of mathematics, others cognitive sciences and so on. It can rely on statistical approaches to give the ability to *learn* from data using two phases. The first one, namely Model Design Phase (Training) consists of estimating a model from data, called observations. The second one is Production Phase where the model is being determined, new data can then be submitted to obtain the result corresponding to the desired task. Handwriting recognition is a good and complex example because two similar characters are never exactly equal. An automatic learning system can be designed to learn to recognize characters by observing *examples*, that is, known characters. According to the information available during the learning phase, it is qualified in different ways. If the data is tagged, it is a supervised learning. We are talking about classification if the labels are discrete, or regression if they are continuous. If the model is learned incrementally based on a reward, it is called reinforcement learning. When data (or *tags*) are missing, the model must use untagged examples that can still provide information. In medicine, for example,

it can be an aid to diagnosis. It is said that learning is semi-supervised. While the labeling of data is partial when a model states that a data does not belong to a class A, but perhaps to a class B or C (A, B and C being 3 diseases for example evoked as part of a differential diagnosis). This is called partially supervised learning. In the most general, unlabeled case, we try to determine the underlying structure of the data: this is unsupervised learning. Automated learning is used for a wide range of applications, such as diagnostic aid [?], outlier detection, missing data detection, fraud detection, financial market analysis [?] and relevant information retrieval from multiple sources (social media) [?], and so on. According to Bernardos & Vosniakos, 2007 [?], we expect future advancements in vision to be largely based on fully trained systems that combine Convolutional Networks (ConvNets) and Neural Networks using reinforcement learning (RNNs) to decide where to look [?]. It is not just about a set of algorithms, but a list of steps to take into account and execute in order to reach an optimal result. Data Acquisition is the first step of this list where the algorithm feeds on input data and where the success of the project is collecting relevant data and in sufficient quantity. Preparation and cleaning of the data is the second step. The third is creation of the model. The fourth step is Evaluation which consists of evaluating trained model on the other (second) set of data. The fifth is Deployment where the model will be deployed in production to make predictions, and potentially use new input data to re-train and improve its model.

However, care must be taken to use an adequate number of neurons and hidden layers and to detect over-learning. Thus, the data is divided into two subsets [?]: the learning set which allows changing the weight of the neural network. The validation set allows verifying the relevance of the network. To have a network that generalizes well, we stop learning as soon as we observe this divergence between the two curves. We can also reduce the size of the network and start learning again. To avoid over-learning, a form of regularization is used. The weight decay method is a regularization technique used to limit over-learning in a neural network.

A number of supervised or unsupervised learning models are reviewed here. Among them, we have *Data Mining*. Essential component of Big Data technologies, large data analysis techniques and recently data smart streaming², it is often defined as the process of discovering new knowledge by examining large amounts of data (stored in warehouses or in streaming) using pattern recognition technologies as well as statistical and mathematical techniques. This knowledge, unknown at first, may be correlations, patterns, or general trends in that data. Experimental data are necessary to verify the correction of the system or the estimation of some difficult parameters to mathematical modeling. While *Text Mining* is a branch of Data Mining that specializes in the processing of text corpora to analyze the content and extract knowledge [?]. It refers to linguistic technologies that make it possible to switch from text (full text) to a digital vector (presence-absence or frequency) [?]. String Mining is the next model that channel exploration usually deals with a limited alphabet of elements that ap-

² <https://www.lebigdata.fr/smart-data-definition-differences-big-data>

pear in a sequence, but the sequence itself can usually be very long. Examples of alphabets may be those of the ASCII character set used in the natural language text, nucleotide bases **A**, **G**, **C** and **T** in DNA sequences or amino acids for protein sequences and examining gene and protein sequences to determine their properties [?]. The model Sequential Pattern Mining consists of discovering unexpected and useful models in data sets. It consists of discovering interesting sub-sequences in a set of sequences, where the interest of a sub-sequence can be measured according to various criteria such as its frequency of appearance, its length and its profit. The extraction of sequential models has many real applications because the data is encoded as sequences in many areas such as bio-informatics such as genomics and Proteomics. We have also Web Mining where, thanks to the huge amount of information available online, the World Wide Web is a fertile ground for research in data mining. Web mining research is at the crossroads of research conducted by several research communities, such as databases, information retrieval and within IA, particularly the sub-domains of learning and natural language processing.

2.2 Disaster & Disaster Management

Over the past two decades, many disasters have emerged in addition to existing disasters, mainly anthropogenic disasters due to globalization, interconnected networks and considerable technological development [?]. Anthropogenic disasters include product forgery, biological threats, terrorism and ecological terrorism. Figure ?? summarizes few examples of disasters with their type, location, year and number of victims.

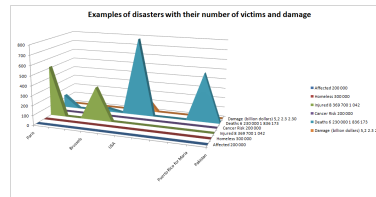


Fig. 1. Examples of Disasters

From these examples [?], in the period from 1980 to 2017, we have 160,000 deaths and 60 million wounded and in the period from 2012 to 2017, we have 32,454 dead, 3,355 injured, 6,639 missing, more than 83,000 hectares burned, 350 homes destroyed and considerable damage just in the past five years [?].

Recent threats from these disasters have reaffirmed the urgency and importance of loss assessment and the need for decision support tools. In order to meet these needs and requirements, different disaster management systems have been studied and developed. Some of the main activities in disaster management are as

follows [?], namely Hazard assessment, Risk Management, Mitigation, Preparedness, Response and Recovery. When disaster strikes, people seek information and ways to provide information to help others. Disasters promote altruism, where people help those who are in difficulty or who suffer from the disaster. Among the most common types of online help in a disaster, they are in the form of information on the safety of people and goods which are more frequent and faster than that of official reports. Google Maps also provides additional help online: a Web application that produces annotable maps. As a result, disaster management research and practice often refers to a formula (??) [?] of the following type:

$$\mathbf{R} = \mathbf{D} * (\mathbf{V} - \mathbf{Res}) \quad (1)$$

Where: R is the risk which is the probability or expectation of a loss; D is the danger which is a condition that threatening the danger. V is the vulnerability which is the extent to which people or things are likely to be affected; and Res is the resources which are the assets in place that reduce the effects of hazards.

While it is not possible to prevent a disaster, the risk of a disaster can be minimized by controlling risk and vulnerability, and improving the capacity of individuals and communities to overcome these risks. Understanding the source of common danger, disasters can vary in terms of types and impacts on human societies [?]. Disasters can be classified into two main categories: natural and anthropogenic. Natural disaster include geophysical, biological and hydro-meteorological disasters. As for anthropogenic disasters, they consist of terrorist acts, simple accidents and technological accidents. Figure ?? summarizes our new classification of disasters.

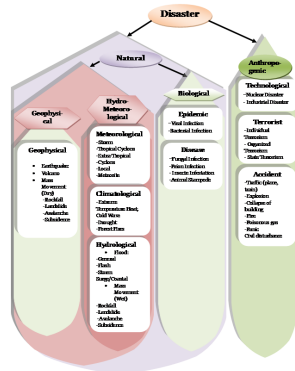


Fig. 2. New Classification of Disasters

Disaster is said technological if it is due to a technical or human failure, occurred in a facility classified as dangerous for the protection of the environment, namely a factory, a depot, a site, a quarry and damaged a large number of real

estate, private or professional. Among these technological disasters, there are nuclear disasters and industrial disasters. Industrial disasters can be the result of negligence or incompetence, or any combination of these factors. These can usually result in more or fewer deaths, injuries or disappearances, as well as significant property damage. These consequences can be immediate or delayed. A nuclear disaster is a serious accidental industrial event, the consequences of which are related to the presence of radioactive materials.

Terrorism is a crime against human life in violation of national and international laws while having a political purpose that would influence or change the policy of a country by terrorizing its civilian population, according to the philosopher J. Derrida . There are four main types of terrorism, namely individual terrorism, provoked by rebels, anarchists or nihilists, organized terrorism, advocated by groups with different ideologies such as the extreme left, the extreme right ..., state terrorism and cyber-terrorism. These different types of terrorism are motivated by four elements, namely Vindictive hatred (hatred usually leads to revenge for the enemy's abuse), deterrence (so that the terrorized population puts pressure on its government), propaganda (to strike the spirits) and provocation (to push a government to overreact). Before introducing the relationship between danger, vulnerability and capacity, it is important to understand the meaning of these terms. Vulnerability refers to the inability of individuals, households and the community to prepare for and respond to hazards. The disaster can be far too serious that, individuals should be unable to act or make the appropriate decisions to solve the problems they face.

The relationship between danger, vulnerability and capacity is generally represented by the formula (??) [?] as follows:

$$\mathbf{Dis} = \mathbf{D} * (\mathbf{V}/\mathbf{A}) \quad (2)$$

Where: Dis is the disaster, D is the danger to the disaster, V is the vulnerability to danger and A is the ability to overcome a disaster.

This equation clearly shows that the risk of disaster for a community increases because of the high probability of a disaster (danger) and its inability to cope with the disaster. However, the use of adequate knowledge, skills and resources for disaster preparedness and response can counteract the overall risk of a disaster.

- Mitigation (before the disaster): is the process of planning and taking long-term risk reduction measures for people and property in the event of a disaster [?]. This eliminates the risks before a disaster occurs.
- Preparedness (before the disaster): is planning how to respond to a disaster [?].
- Response (in case of disaster): consists mainly of conducting an assessment of the situation and implementing the strategies developed during the preparation phase [?]. It responds to the needs of the victims and take the necessary steps such as public warning, safe evacuation, management of persons or

victims, measures to restore the basic structure of life and short-term plans to restore the normalcy of daily life [?].

- Recovery (after a disaster): is about ensuring that all community activities and systems return to the normal situation [?]. These actions can be classified into two main categories, namely short-term recovery and long-term recovery.

Disaster management is roughly categorized into two types [?], namely structural measures (SM) and non-structural measures (NSM). SM can reduce physical damage directly related to disasters (eg. embankments, seismic buildings and evacuation sites). However, SMs are expensive, time-consuming, and can not necessarily mitigate the damage caused by unexpected events. On the other hand, NSM are based on knowledge or practice; it can provide people with motivation and knowledge of disaster management at a relatively low cost (eg. policy and law development, evacuation drills and disaster education).

Several disaster management approaches [?] have been observed in recent decades. The most classical approach for disaster management, depending on the geography, is called geo-medical perspective; it determines the probability of disasters, the consequences for public health and the type of medical interventions to be implemented to reduce human losses. The strategic-operational disaster approach examines how individuals, organizations and communities develop their skills in response to unexpected events such as organizational resilience in disaster preparedness and management and also in the institutional aspect. The communicative approach of disaster management underscores the growing importance of serious agency consideration in disaster management planning (eg. political communication and concurrent understanding of the disaster). Finally, the community-based approach to disaster management examines familiar knowledge such as indigenous knowledge, familiarity and specialized knowledge. Familiar knowledge giving rise to the formation of an imaginary risk is an important factor in increasing the vigilance of a community in the face of disasters.

1 Classical Disaster Management Model

It is possible to reduce the disaster risk, as with earthquakes, by preventive actions, starting with the information of the citizen by teaching them the attitude to adopt before, during and after disaster, then reducing the vulnerability seismic of buildings, which can limit the damage. This solidarity action is illustrated and modeled in the following section by the collaboration between all the volunteers. The field and profession of disaster management is already evolving into a more collaborative enterprise. It is modeled by the following equation:

$$CE = VC - D \text{ where } VC = VC_i \text{ with } i \in [1, N] \quad (3)$$

Where:

VC is the set of Volunteers Contents,
 D is the set of Redundant Contents of Volunteers and
 VC_i is the i^{st} Volunteer Content.

Volunteers provide leading-edge capacity and links to community resources. The mobilization of organizational and individual volunteers also serves a social and psychological purpose: it brings communities together and gives them a sense of effectiveness. More adaptive management would promote organizational learning and facilitate adaptation and improvisation.

2 Computer Disaster Management Model

The "maturity" of an information system design to be effective in the humanitarian context is assessed by its degree of integration of humanitarian values. It must, before all, encourage the development of disaster management skills with, for example, disaster education or simulations [?]. The use of simulation in the context of humanitarian logistics can help decision makers simulate distinct modes of transport in order to understand, analyze and improve the allocation of resources as well as the delivery time of goods. Floods caused by brief and abundant showers [?] were analyzed and other factors were identified to find relatively inexpensive solutions to the traffic problems generated by the floods. These spontaneous floods, caused by brief and abundant showers, are traditionally attributed to loss of infiltration by urban concrete, a secular drainage system and clogged watercourses.

3 Social Network Model for Disaster Management

Following the increase in social networking sites and applications, it has attracted the attention of researchers and practitioners. On the one hand, practitioners are developing skills to make better use of this online communication technology and, on the other hand, researchers are developing models to examine various aspects of the use of social networking in different environments and disaster management. It is important to note that social media is not considered a substitute for traditional media such as television, radio and newspapers but just that their importance is undeniable as an additional means of communication [?], especially when sharing information is essential - as in disaster management.

2.3 Social Networking

Social Networking plays a very important role in disaster management. Collecting and classifying the content (information and images) posted on social networking platforms, to help humanitarian organizations to become aware of the situation and to launch relief operations, understanding these voluminous and high velocity data is a difficult task which requires the following Crowdsourcing, Crowdtasking and Collaborative Disaster Management.

– Collaborative Disaster Management

The catastrophe is a situation that requires quick and effective collaborative decisions, with a high degree of uncertainty and complexity [?].

– **Crowd-sourcing and Crowd-tasking in Disaster Management**

Crowd-tasking describes the management of volunteers applying new media and the Crowd-Tasking System (CTS) is defined as combination of technical solutions (crowdtasking applications, CTA), task managers, task management software in the situation room and volunteers themselves [?].

The rise of social networks has enabled ad hoc citizen groups to organize large-scale activities in a flexible way. In post-disaster relief operations, it is assumed that all fixed cellular infrastructures do not be functional. The adaptability of Peer-to-Peer (P2P) networks must be exploited to respond to the characteristics of disaster situations. P2P networks have been used to interconnect field workers by just one active connection between a peer and the control room to perpetuate the disaster management system. Thus, the benefits of applying the P2P paradigm have applied to the Geo-Collaboration for Disaster Management as an alternative to the most common Client/Server approaches of GIS and propose to interconnect mobile operators through P2P networks. Mobile P2P systems are a relatively new approach to combine P2P concepts with mobile computing technologies such as WirelessLAN, Bluetooth and telecommunication networks such as GSM, GPRS or UMTS. Mobility support is becoming an increasingly important research topic in the design of P2P systems, especially in Disaster Management. As a result, users' flexibility and independence can also be increased in a physical way, so that they can interact with P2P-based mobile anytime, anywhere mostly in Disaster Management. The study found that New Trends in Scientific Research in Disaster Management focus on two broad trends, namely [?] prediction and monitoring for early warning, and Relevant information retrieval and situational awareness classification.

Models for retrieving information from multiple sources

In a bilingual recovery, comparable document-aligned data [?] is used to easily learn patterns of common topics or word concepts. However, the preparation of comparable documented data requires a lot of human involvement and time. Therefore, such approaches are not appropriate when information about an ongoing event should be retrieved quickly. The work of Joseph A. Shaw and Edward A. Fox [?] focuses on methods for recovering separately from different sources and then combining them using data fusion techniques. Multi-view recovery based on deep learning that attempts to learn document embedding in a common space, where differences between different data sources would not exist. Unfortunately, this approach has low performance scores for all the methods used: indicating that the problem is difficult and requires better methods. According to Cynthia D. Balana [?], Google Crisis Response's (GCR) National Risk Management and Disaster Response Board (NDRRMC) had integrated Facebook and Twitter into its surveillance system and website for a quick response.

Whether it's Twitter, Facebook, Viber, Messenger, any forum or any thing in the Web, these are platforms where people often express emotions. Emotions are detected using four types of features, namely: interjections, blasphemy, emoticons, and the general feeling of the message.

Table 1. Comparative table of all techniques and methods used in Models for retrieving information from multiple sources.

Retrieving Information Approaches	Exigences	Real-Time	Multiple Sources	Recuperation Methods
[?]	/	/	Surveillance System Facebook and Twitter	/
[?]	Requires a lot of human involvement and time	Not appropriate for real-time approach	/	Learning patterns with comparable documents
[?]	/	/	Recovering separately different sources	Combining using data fusion techniques
[?]	Low performance scores	/	/	Multi-view recovery based on deep learning
Proposed Approach	Annotating manually information	Appropriate for real-time	Multiple sources (Facebook Twitter)	Learning with Neural Network

Interjections, blasphemies and emoticons are widely used by individuals to convey emotions such as anger, surprise, happiness, etc. To identify these three types of features, a combination of POS tags is used in the English tagger (which contains tags for interjections, emoticons, etc.), compiled lists of interjections and blasphemies on the Web for English and regular expression patterns for emoticons.

Table ?? shows a comparison of all techniques and methods used in Online Listening and/or Monitoring.

Our proposed model is based on a semi-supervised inductive technique to use unlabeled, multi-source data, which is often abundant during a crisis event, with less data previously labeled than previous events (*Table ?? and Table ??*).

Online Listening and Monitoring

Contents were collected from all online channels tracked automatically by the Online Listening Tool, namely Radian6 [?], from websites to social media: Twitter, Facebook, LinkedIn, Instagram, Google+, Youtube and so on. Radian6 tool or any its competitor ³, such as Awario, Brand24.com, Brandwatch, Mention, Keyhole, Socialert.net, SocialPilot.co, Simplify360, etc. of Online Listening or Monitoring tools, is used to collect information, using the search keywords 'Boumerdes' and 'earthquake' for the Boumerdes earthquake and the 'Algiers' and 'flood' keywords for the Algiers floods. Online listening tools provide the model, which reasonably represents the essentials, namely:

1. Harvesting contents: (such as conversations at the social media, news or any information in the Web);

³ <https://www.quora.com/What-are-the-top-competitors-to-radian6-I-am-looking-for-alternate-measuring-tool-to-use-with-my-clients>

Table 2. Comparative table of all techniques and methods used in Online Listening and/or Monitoring.

Online Monitoring Approaches	Lis- ing and Monitoring Tool	Online Listen- ing and Moni- toring Projects	Methodology to Listening and Monitoring Projects	Approach Used	OSN
[?]	Twitter APIs for tweets and Radian6 for Facebook messages	Based on keywords- Annotated manually , harvesting contents, cleaning them and extracting relevant information	Twitter & Facebook		
[?]: our previous work	Twitter for tweets and Radian6 for Facebook messages	APIs Based on keywords- Annotated manually , harvesting contents, cleaning them and Learning Relevant information with neural network, in <u>Real-time</u>	Twitter & Facebook		
Proposed Approach	Radian6 for all contents	Based on keywords- Annotated manually messages, , harvesting contents, cleaning them and Learning Relevant information with neural network, in <u>Real-time</u>			

2. Cleaning the data of duplication and replication content: eliminating, from the content, any dubbed information like re-tweet, and any harmful or redundant information;
3. Enabling relevance: thanks to the neural learning, obtaining of relevant information using the corpus. This corpus is the annotated manually content which is realized by volunteers;
4. Analyzing the results: During this stage, the verification and analysis of the results is carried out in order to ensure adequacy so that it is obtained to alert the disaster managers.

It is more the limitation of freedoms by monitoring the attitudes of entire populations that I consider the main disadvantage of the online tools of listening and monitoring. As for the benefits, they are huge. For example, finding some details of an ultimate discussion is one of them. Being able to predict and prevent the outbreak of a disaster and make certain decisions that could save lives, has no commercial value. It is above all a value of great morality.

Table 3. Comparative table of all techniques and methods used in Models for retrieving information from multiple sources.

Retrieving Information Approaches	Exigences	Real-Time	Multiple Sources	Recuperation Methods
[?]	/	/	<i>Surveillance System / Facebook and Twitter</i>	
[?]	<i>Requires a lot of human involvement and time</i>	<i>Not appropriate for real-time approach</i>	/	<i>Learning patterns with comparable documents</i>
[?]	/	/	<i>Recovering separately from different sources</i>	<i>Combining using data fusion techniques</i>
[?]	<i>Low performance scores</i>	/	/	<i>Multi-view recovery based on deep learning</i>
Proposed Approach	Annotating manually information	Appropriate for real-time	Multiple sources (Facebook Twitter)	Learning with Neural Network

2.4 Disaster Education

Disaster education plays a key role, first, in helping citizens to become aware of the disaster dangers, and then, in encouraging them to develop skills in disaster management. This environment is designed to support an introductory traineeship in disaster management for trainees and future disaster managers. Community members usually share their knowledge and help one another in disaster preparedness. Evacuation drills often take place in schools, businesses and communities [?], [?], [?], [?], [?] and [?].

2.5 Related works

In 2009, Tobita et al (2009) [?] have initiated by introducing an Integrated disaster simulator using Web GIS and its application to community disaster mitigation activities. In 2012, Fischer et al (2012 [?]) introduced Dark Tourism, Penumbra Tourism, Mixed Realistic Games or Alternative Realistic Games (ARG). In 2015, Tsai et al (2015) [?] introduced a flood protection computer game for disaster education. Followed by Mitsuhashi et al (2015) [?] developed a Game-based evacuation drill using real world edutainment. Finally, in 2016, Kawai et al (2016) [?] developed a Tsunami Evacuation Drill System using Motion Hazard Map and Smart Devices. While Power and Robinson (2016) proposed Comparing Felt Reports and Tweets About Earthquakes.

3 Automated Learning Environment in Disaster Management

3.1 Automated Learning Environment: modeling overview

Artificial Neural Networks (ANN), inspired by biological neural networks, are like the brains of animals. Such systems *learn* to perform tasks by considering examples, usually without being programmed, with task-specific rules. A neural network is based on a collection of connected units or nodes called neurons, which model the neurons of a biological brain. Each connection, like the synapses of a biological brain, can transmit a signal from one artificial neuron to another.

Learning by retro-propagation is a method of gradient descent: in a given iteration, the direction of the search is given by the negative gradient of energy, while the pitch in this direction is given by a constant (velocity learning) chosen by the user. However, it is well known in the optimization literature that pure methods of gradient descent tend to be very inefficient. While keeping its great advantage, namely obtaining a speed of convergence faster around the global optimum and, at the same time, a convergence accuracy that can be higher.

The feedforward artificial neural networks (ANN) are currently used in various applications with great success. Their first advantage is that they do not require a userspecific problem solving algorithm (as in the case of classical programming), but they learn from examples, much like human beings. Their second advantage is that they have an inherent capacity for generalization. This means that they can identify and respond to patterns similar to but not identical to those with which they were formed. On the other hand, the development of a feedforward ANN model also poses some problems, the most important being that there is no guarantee that the model will work well for the problem to be solved [?].

The entire ANN modeling procedure has been studied with the aim of introducing systematic methods leading to always efficient ANN models, namely the collection of learning data, preprocessing and post-processing of data, different types of activation functions, initialization of weights, learning algorithms and error functions. Although all of these factors affect ANN's performance, increased attention has been focused on finding the best architecture. This is justified not only by the fact that it is directly related to the performance of the model, but also by the fact that there is no theoretical background on how this architecture will be found or on its appearance [?]. The most typical method followed is a repetitive trial and error process, in which a large number of different architectures are examined and compared to each other. Therefore, this process is time consuming and relies mainly on the experience and intuition of the human expert, which implies a high degree of uncertainty. Nevertheless, we cite four different approaches, namely:

- The empirical or statistical methods used to study the effect of internal parameters of an ANN and choose appropriate values depending on the performance of the model [?],

- The hybrid methods such as fuzzy inference [?],
- The constructive and / or pruning algorithms [?] and finally
- The evolutionary strategies.

The most systematic and general of these methods uses the principles of Taguchi's conception of experiments [?]. The best combination of the number of hidden layers, the number of hidden neurons, the choice of input factors, the parameters of the training algorithm, etc. can be identified with these methods even though they are usually case-oriented.

In the hybrid methods [?], the ANN can be interpreted as an adaptive fuzzy system or can operate on fuzzy numbers instead of real numbers. In the constructive and / or pruning algorithms that, respectively, add and / or delete neurons of an initial architecture using a previously specified criterion to indicate how the ANN performance is affected by the changes [?]. The basic rules are that neurons are added when training is slow, and that neurons are suppressed when a change in the value of a neuron does not match at a change in the response of the network or when the weight values associated with this neuron remain constant during a large number of training periods.

In the evolutionary strategies, the topological space is explored by varying the number of hidden layers and hidden neurons by applying genetic operators and evaluating different architectures according to an objective function. Since the constructive and pruning algorithms are essentially gradient descent methods, their convergence towards the global minimum is not guaranteed and can therefore be trapped at a local minimum close to the point of the search space from which algorithms has been launched.

An artificial neuron that receives a signal can process it and then report other artificial neurons connected to it. In current implementations of neural networks, the signal at a connection between artificial neurons is a real number and the output of each artificial neuron is computed by a nonlinear function of the sum of its inputs. The connections between artificial neurons are called *edges*. Neurons and artificial edges usually have a weight that adjusts as learning progresses. The weight increases or decreases the signal strength during a connection.

Deep learning allows computer models composed of multiple layers of processing to learn data representations with multiple levels of abstraction: this has greatly improved the state of the art in speech recognition, visual recognition and objects, object detection and many other areas such as bioinformatics, Sequential Pattern Mining or genomics [?].

3.2 Configurations of the Neural Network Parameters

Problem: how to determine the coefficients w_0 , w_1 and w_2 ? To the Error function: If $y = 1$, we want to have $p(x)$ as large as possible. We define the error in this case by $-\ln p(x)$.

Symmetrically, if $y = 0$, we want to have $p(x)$ as small as possible. The error is then $-\ln(1 - p(x))$. So, the general formula (??) is :

$$error = -y * \ln(p(x)) - (1 - y) * \ln(1 - p(x)) \quad (4)$$

Figure 5.1 shows that a given neuron in a layer k consists of elements such as input flux of all precedent neurons and output flows towards all the following neurons with their respective weights.

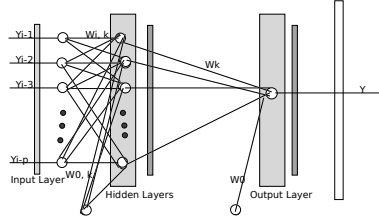


Fig. 3. Neural Network Structure.

Figure ?? shows Feedforward Neural Network Architecture.

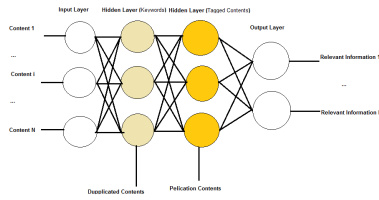


Fig. 4. Feedforward Neural Network Architecture.

Minimization of the error: Once an error function has been defined, the problem of learning becomes an optimization problem: find the coefficient vector w^* which minimizes the error. In the case of logistic regression, this vector is unique because the error function is convex. The solution vector w^* can be obtained by an iterative algorithm.

Probability of classifier error: Once the optimum w^* coefficient vector is determined, a program (classifier) is available to classify a new individual. To estimate the error probability of the classifier, it is necessary to have an independent test set.

The basic version of frequent pattern extraction is used to search a table in a Content Set whose values are Booleans indicating the presence or absence of a property. O is a finite set of objects. P is a finite set of properties. $\mathcal{R}$ is a relation on $O \times P$ which makes it possible to indicate if an object x has a property p (noted $x\mathcal{R}p$) or not. For example, in the case of basket analysis in a supermarket, O is the set of purchase transactions, P is the set of items and $\mathcal{R}$ is the relationship indicating whether an item is purchased in transaction t .

3.3 Manipulating of the Automated Learning Environment

We used the Model-View-Controller (MVC) model for event management design. We have also extended it to data mining specifically to the Sequential Model Exploration that we can apply to Marketing for example.

Figure 5.2 shows the launching of Feedforward Neural Network (Automated Learning Environment).

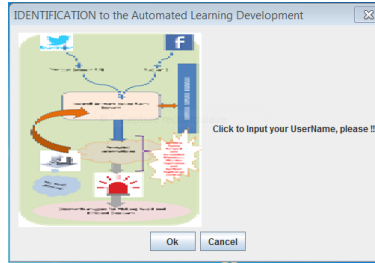


Fig. 5. Identification in the Automated Learning Platform

We used the Apriori algorithm used in Knowledge Discovery in Databases (KDD). It has been improved by Savasere [?]. We have adapted it to streaming in order to use the data of social networks. We have also used and adapted the DIC algorithm of [?] to streaming. Figure 5.1 shows the Automated Learning Environment launched.

3.4 Sequential Pattern Mining: algorithms implementation

A pattern of a formal Content Set $(O; P; R)$ is a subset of P . The set of all the motives of a base is therefore the set of parts of P , denoted 2^P . We say that an object $x \in O$ has a pattern m if, $\forall p \in m; x \mathcal{R} p$. For the example of the content set, we have: Size pattern $0 = \emptyset$; ($C_5^0 = 0$ reasons). Size reasons $1 = a; b; c; d$ and e , which will be noted, for simplicity, $a; b; c; d$ and e . ($C_5^1 = 5$ reasons) Grounds of size $2 = ab; ac; ad; ae; bc; bd; be; cd$; this, of ($C_5^2 = 10$ patterns) Size reasons $3 = abc; abd; abe; acd; ace; ade; bcd; ecb; bde; cde$ ($C_5^3 = 10$ units) Size reasons $4 = abcd; abce; abde; acde; bcde$ ($C_5^4 = 5$ patterns). Size $5 = abcde$ ($C_5^5 = 1$ patterns). In the previous formal set, x_1 has the patterns: $at; c; d; ac; ad; cd$ and acd . Among the global set of 2^P patterns, it will look for those that appear frequently. Let m_1 and m_2 be two patterns. An association rule is an implication of the form: $m_1 \implies m_2$ where $m_1, m_2 \in 2^P$; and $m_1 \cap m_2 = \emptyset$; The rule $m_1 \implies m_2$ is verified in data set E with a support s , where s is the percentage of objects in E containing $m_1 \cup m_2$. Let $m \in 2^P$ be a pattern. The support of m is the proportion of objects in O that have the pattern: Support: $2^P \implies [0; 1]$ $m \implies \text{Support}(m) = (|f(m)|)/(|O|)$ For example in the previous database, we have: Support $(a) = 3/6$; Support $(b) = 5/6$; Support $(ab) = 2/6$; Support $(\emptyset)$

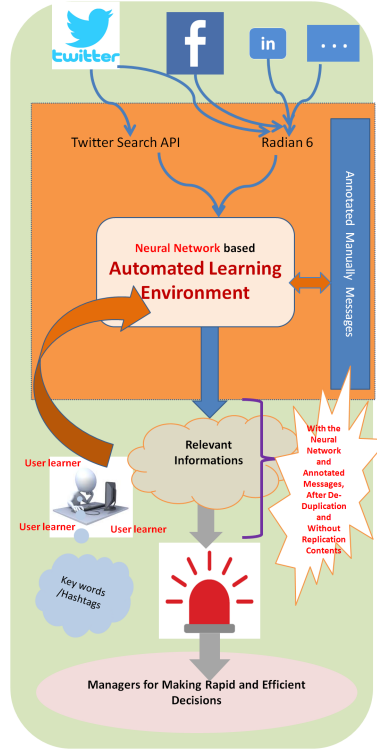


Fig. 6. The Automated Learning Environment

$= 1$ and $\text{Support}(P) = 0$. The support is decreasing by $(2^p; \subset)$ in $([0; 1]; \leq)$. In other words, if m is a sub-pattern of m' ($m \subset m'$) then $\text{Support}(m) \geq \text{Support}(m')$. The medium measures the frequency of a pattern: the higher it is, the more frequent the pattern. Frequent patterns of non-frequent patterns can be distinguished using a threshold σ_s [?]. Streaming is a way of broadcasting and reading streaming content, which is widely used on the Internet. It is opposed to the file download which requires to recover all the data of a file. However, playing streaming content involves being connected to an Internet server. Smart Data⁴, a different concept of Big Data, is based primarily on real-time data analysis. This term refers to an approach to data analysis that directly analyzes the data at the source, without the need to transmit it to a centralized system.

Dynamic counting algorithm: The DIC algorithm, proposed by Brin et al. (1997) [?], to reduce the number of runs of the database, is suitable for streaming. Thus, DIC, which partitions the database into blocks of M transactions, and adapted to work with blocks of M contents. During the computation of the k -itemsets supports, after the search of a partition of size M of D , we verify the

⁴ <https://www.lebigdata.fr/smart-data-definition-differences-big-data>

k-itemsets candidates who have already reached the minimum support, DIC then uses them to generate candidates of size $(k+1)$, and starts counting their supports. Thus the supports of candidates of different sizes are calculated during the same course of D. As the number of scans in the database decreases, there is only one passage of the content, whereas DIC considers candidate itemsets of different sizes simultaneously. This poses the problem of storing the candidate itemsets processed simultaneously and the cost of calculating the candidates' media which is greater than for the Apriori algorithm [?].

AprioriTID algorithm: The Apriori algorithm [?], requiring N passages on the database, a possible optimization consists of generating in memory, during the first pass, the identifiers (TID) of the transactions for each 1-itemset (together pattern size 1) frequent. This algorithm is also suitable for streaming. The TID lists corresponding to each k-itemset of the K contents are kept. The calculation of a k-itemset is always done from the two $(k-1)$ -itemsets containing one less element, but the counting is done simply by intersection of the two TID lists of the two $(k-1)$ -itemsets source. We build the list of TIDs after determining the frequent 1-itemsets, which is more efficient in streaming. When reading the first content, this eliminates the infrequent products and thus reduces the lists of TID in memory. Generate TID lists in parallel memory as soon as the first content is more efficient. Here is this Apriori reference algorithm adapted to streaming.

Apriori Algorithm: Algorithm 1 shows the AprioriTID algorithm which is the algorithm adapted to streaming.

Algorithm 1: AprioriTID algorithm

```

1: begin
2: Require: Content of Social Networks D, Minimum Support Threshold  $\sigma$ ;
3: Ensure: Set of frequent items;
4: input ( $a$  content  $V(w_1, w_2, \dots, w_i, \dots, w_n)$ ) and  $\sigma$ 
5: initialization  $i \leftarrow 1$ ;
6: initialization  $C_i \leftarrow 1$  set of size 1 patterns (one item);
7: while ( $C_i \neq \emptyset$ ) do
8: Calculate the Support of each pattern  $m \in C_i$  in the content set:
9:  $F_i \leftarrow \{m \in C_i \mid \text{support}(m) \geq \sigma\}$ ;
10:  $C_{i+1} \leftarrow$  all possible combinations of  $F_i$  patterns of size  $i + 1$ ;
11: incrementation  $i \leftarrow i + 1$ ;
12: endwhile
13: output  $\bigcup (i \geq 1) F_i$ 
14: end

```

1. The application of the algorithm on the basis of the given contents read in streaming given in example with $\sigma = 0.25$ happens as follows. Supports: 3/6, 5/6, 5/6, 1/6 and 5/6. Hence, $F_1 = \{a; b; c; e\}$ (no frequent reason will contain d).
2. Generation of size 2 candidates. Combine 2 to 2 F_1 size 1 candidate. So, $C_2 = \{ab; ac; ae; bc; be\}$ and Supports are 3/6, 2/6, 4/6, 5/6 and 4/6. $F_2 = C_2$: all C_2 patterns are common.
3. Generation of size 3 candidates: Combine 2 to 2 F_2 size candidates (and consider only those who give size 3 patterns). So, $C_3 = \{abc; abe; ace; bc\}$ and Supports are 2/6, 2/6, 2/6 and 4/6. $F_3 = C_3$: all C_3 motifs are common.
4. Generation of size 4 candidates. So, $C_4 = \{abce\}$ and Support is 2/6.
5. Generation of size 5 candidates: $C_5 = \emptyset$; So, $F_5 = \emptyset$;
6. The algorithm then returns all of the common patterns, namely: $F_1 \cup F_2 \cup F_3 \cup F_4$.

Among the goals of the optimizations this algorithm is to facilitate streaming playback with a single read with sufficient computation and storage. The threshold σ is set by the analyst. This can follow an iterative approach by setting a threshold at the start and, depending on the result, change the threshold value. The algorithm proposed by Savasere [?] solves the memory space problem of the previous algorithm. The advantage of this algorithm is that it requires only one reading at most.

3.5 Disaster Education: overview

This environment is designed to support an introductory traineeship in disaster management for citizens, trainees and future disaster managers. The trainee can use this tool in the following modes:

- 1 Novice learning environment. To work with this tool, the trainee can use a complete set of automated design and learning tools. One of the functions of the environment is to display visualizations of the trainee's programs. With the help of the environment, the trainee can observe various programs "at work", experimenting them and gradually learning from his experience, observations and mistakes.
- 2 Learning environment for beginners. At any point in the job, the trainee can ask this environment to generate (move on) the next step: a new concept or a new build, a build example, a test, an example of a problem solver or a problem to be solved. This tool analyzes domain knowledge and the trainee's model and provides both the optimal stage and a list of all relevant operations (ie, ready to be applied). The trainee who is not satisfied with the optimal operation suggested by the system can choose any appropriate operation using adaptive hierarchical menus.
- 3 Online manual rehearsal. At any time during the work, the trainee has a menu to access all previous courses: presentation of any previously learned concept, demonstration of all the examples learned and analysis of any problem explained or resolved. This mode provides access to the material learned from the course as a reference, thereby supporting example-based online help.

4 Manipulating of the Automated Learning Environment

The trainee, once registered, with his own username and password, and thus defines his own learning environment, can thus launch the learning environment, already configured, to extract content from social media, analyze it. The keywords and manually annotated contents are already operational: they can view with the "Viewing" menu. Listening and Monitoring is programmed to "All the Web". Processing is set to "Streaming", the alert to "Automatic" and Learning to "Neural Learning".

The novice can simply follow the progress of this treatment by viewing the relevant content, the applied content and the replication content using the "Viewing" menu.

In the event that he wants to participate (intervene) au processus, he can, for example, change the type of Learning with the "Learning" menu, change Listening and Monitoring, the treatment or the alert.

It can also intervene by adding other keywords in the "Updating Files / Updating Keywords" menu. He can also modify the alert to make it manual: thus, it will be up to him to alert.

If he is interested in understanding how this tool works, he selects "Content Treatment/Treatment" menu. Thus, it will be up to him to insert a text in the "Updating Files / Inserting Content" menu.

Thus, he can follow his treatment with the "Content management" menu for the number of words retained after processing his message, obtain content sender or the number of occurrences that his message contains. It can know whether it is accepted as relevant, whether it is rejected.

The trainee can, at any time, review all of his work in order to be able to correct his own errors in the future.

This Automated Learning Environment leads, at any time, to visualize, thanks to "Viewing" the different

used available keywords, manually annotated contents, relevant contents, duplicate contents and replication contents.

It permits to work in Streaming or just by providing a message (especially for novices) in "Content Treatment".

In the case of Streaming, we have to specify, in "Listening / Monitoring", whether it is from Twitter (Twitter is chosen optionally), from other Social Networks or from all the Web.

On the other hand, for the novices, one can enter the content, even update Keywords, in "Updating Files". We can satisfy ourselves with an automatic (optional) or manual alert in "Alert". With "File / New", we start a new learning: just enter "Content File" and validate. We will still have this automated learning. We have "Data Mining", "Neural Learning" and "Deep Learning". "Data Mining" includes "Sequential Pattern Mining", "Text Mining" and "Web Mining", which are from the same family, according to Bharanipriya et al,(2011) [?]. "Neural Learning" consists of "Relevant Information Retrieval" [?] and "Deep Learning" is still under development.

5 Discussion about the Automated Learning Environment

Interjections, blasphemies and emoticons are widely used by individuals to convey emotions such as anger, surprise, happiness, etc. To identify these three types of features, we used a combination of part-of-speech (POS) tags in the tagger, compiled lists of interjections and blasphemies on the Web and regular expression patterns for emoticons [?].

Messages from Algeria's target regions are checked for frequency bursts of disaster keywords and processed to identify evidence of a disaster. The advantage of our Automated Learning Platform is that it relies on evidence from social networks' first-hand reports. This multi-view template associates an input document vector with a multi-view space, based on neural network, whose contextual information is available.

Let $\hat{H}$ be the non-empty set ($\hat{H} \neq \emptyset$) of the terms that we have compiled (for example, terrorist act, explosion, earthquake, etc.) that make up the set of keywords or hashtags for natural and anthropogenic disasters.

Let $e_i \in C_n$, with $i \in [1, N]$, the incorporation of a content of the source message i relevant for, at least, a keyword or a hashtag $h_j \in \hat{H}$ with $j \in [1, M]$. C_n is the set of n contents.

We want to learn a generic space with the neural network

$$\hat{E} = \{e_k \text{ with } k \in [1, K]\} \quad (5)$$

which normalizes the differences:

$$\hat{E} = [E - \hat{R} - \hat{D}] \quad (6)$$

where: $\hat{R}$ is the set of retweets, $\hat{D}$ is the set of duplicate contents, and helps in better recovery of target content.

Thanks to the neural network, the transformation of e_i into e_k can be explained by:

$$e_i \rightarrow e_k = \{e_i, e_i \text{ is relevant for } h_j, e_i \notin \hat{R}, e_i \notin \hat{D}, h_j \in \hat{H}, j \in [1, M]\} \quad (7)$$

where $\hat{D}$ and $\hat{R}$ represent respectively the set of duplicate contents and retweets. The objective is then to maximize the size K of the set $\hat{E}$.

In order to show the effectiveness of our model, we examined a specific event - the Boumerdes earthquake in May 21st, 2003 and the Algiers flood in November 10st, 2001 - and post-event messages on social networks.

Tweets were collected using the Twitter Search API and posts from other social networks were collected using the Radian6 tool. Both use the search keywords "Boumerdes" and "earthquake" for the Boumerdes earthquake and the "Algiers" and "flood" keywords for the Algiers floods.

Thanks to the neural network and after processing the content removing any inconsistencies such as punctuation, special characters, de-duplication and replication content [?], we have obtained figure 5.1 and figure 5.1, which present the

number of relevant documents found for, respectively, each keyword or hashtag [?].

Standard relevance queries and judgments: we identified a set of disaster-specific information needs, as proposed in [?]. It's a set of hashtags and keywords for Twitter and other social networks respectively. Data collection and filtering are at the heart of disaster management using social media. The algorithms used to warn and alert depend on the messages published on the social media and their quality. Thus, every effort must be made to maximize the number of contents relevant to the disaster and to remove non-informative messages as proposed in [?]. These contents are manually tagged (annotated) to remove those not related to the disaster (non-informative messages). Figures 5.1 and 5.1 present the number of relevant documents found for, respectively, each keyword or hashtag from the datasets.

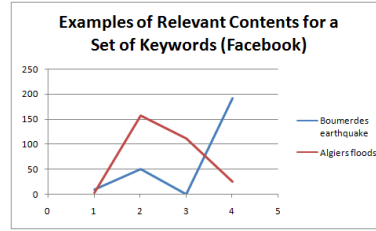


Fig. 7. Examples of Relevant Contents for a Set of Keywords

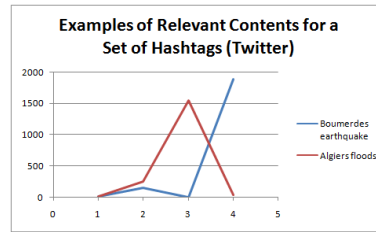


Fig. 8. Examples of Relevant Contents for a Set of Hashtags

This content obtained, manually labeled (annotated) and cleaned from non-informative messages, will be used, in this space based neural network, to analyze live new messages through the learning allowed by this space based neural network, enriched by relevant messages and annotated manually.

The overall effectiveness of a large-scale disaster emergency response is a difficult entity to measure. Every disaster is different and every answer is different. In fact, similar events offer totally different results in their aftermath. For ex-

ample, a category two hurricane can inflict much more damage than a category four, depending on the location of the impact. In addition, the response to an emergency rural and mountainous region of Kabylia, in Algeria, for example, will be totally different from the emergency response to a flood in capitals of major Algerian cities, as Algiers, Oran or Constantine. To effectively measure the overall response to these events, appropriate measures must be used and interpreted appropriately.

6 Conclusion and Perspectives

We have discussed in this paper everything related to an automated learning environment such as different models of automated learning, neural network based automated learning and different potential applications. Concluding, we presented our automated learning environment, fundamentally personal, social, distributed, omnipresent, flexible, dynamic and suitable for disaster education. It is an extension of the real-time alert model used for the management of natural or anthropogenic disasters [?]. This automated learning environment is based on a semi-supervised inductive technique of using abundant and unlabeled data, especially in times of crisis that we have extended to data mining, specifically to the sequential model exploration applied to marketing. We used the Apriori algorithm [?] used in Databases for Knowledge Extraction (KDD). It has been improved by Savasere [?]. We have adapted it to streaming. We have also used and adapted the DIC algorithm of [?] to streaming. This is just the beginning of this automated learning environment. It remains to integrate all other models of automated learning, starting with Text Mining to be able to well process all texts from the content. We must also integrate Web Mining with multiple sources, etc. Thus, we will put at the disposal of users, such as disaster managers, a fairly rich and complete environment for training or disaster management.

References

1. Bouhadad Y. , Nour A., Slimani A., Laouami N. & Belhai D., (2004), "The Boumerdes (Algeria) earthquake of May 21st, 2003 (Mw = 6.8): Ground deformation and intensity," *Journal of Seismology*, pp. 497–506, 2004.
2. Asghar S., Alahakoon D. and Churilov L., (2005), "A Dynamic Integrated Model for Disaster Management Decision Support Systems," *In International Journal of Simulation: Systems, Science & Technology*, 6(10), 2005.
3. Imran M., Castillo C., Lucas J., Meier P. and Vieweg S., (2014), "AIDR: Artificial intelligence for disaster response," *in Proceedings of the 23rd International Conference on World Wide Web*, ACM, pp. 159-162, 2014.
4. Imran M., Elbassuoni S., Castillo C., Diaz F. and Meier P., (2013), "Extracting Information Nuggets from DisasterRelated Messages in Social Media," *Proceedings of the 11th International ISCRAM Conference – Baden-Baden, Germany, May 2013*.
5. Abouelhoda M. and Ghanem M., (2010), "String Mining in Bioinformatics," *In Gaber M., Scientific Data Mining and Knowledge Discovery*, Springer, 2010.

6. Kawai, J., Mitsuhashi, H. and Shishibori M. (2016), "Tsunami Evacuation Drill System using Motion Hazard Map and Smart Devices," In 3rd International Conference on Information and Communication Technologies for Disaster Management (ICT-DM), pp. 13-15.
7. Mitsuhashi H., Sumikawa T., Miyashita J., Iwaka K. and Kozuki Y. (2015), "Game-based evacuation drill using real world edutainment," In Interactive Technology and Smart Education, vol.10, No 3, pp. 194-210.
8. Fischer J., Jiang W. and Moran S., (2012), "Atomic Orchid: A mixed reality game to investigate coordination in disaster response," In Proceedings of the 11th International Conference on Entertainment Computing (ICEC 2012), pp. 572-577.
9. Tsai M.H., Chang Y.L., Kao C. and Kang S.C., (2015), "The effectiveness of a flood protection computer game for disaster education," Visualization in Engineering, vol. 3, No 1, pp. 1-13.
10. Tobita J., Fukuwa H. and Mori M., (2009), "Integrated disaster simulator using Web GIS and its application to community disaster mitigation activities," In Journal of Natural Disaster Science, vol. 30, no. 2, pp. 71-82.
11. Power R. and Robinson B., (2016), "Comparing Felt Reports and Tweets About Earthquakes," In 3rd International Conference on 'Information and Communication Technologies for Disaster Management (ICT-DM)', Vienna, Austria.
12. Lagmay A. M., Mendoza J., Cipriano F., Delmendo P. A., Lacsamana M. N., Moises M. A., Pellejera N., Punay K. N., Sabio G. Santos L., Serrano J., Tanisa H. J. and Tingin N. E. R., (2017), "Street floods in Metro Manila and possible solutions," In Journal of Environmental Sciences.
13. Massegia F., Teisseire M. and Poncelet P., (2005), "Sequential pattern mining: A survey on issues and approaches," Encyclopedia of Data Warehousing and Mining, 2005.
14. Balana C. D., (2012), "Social media: Major tool in disaster response," Inquirer Technology, June 15th, 2012.
15. Bouzidi Z., Boudries A. and Amad M., (2018), "A New Efficient Alert Model for Disaster Management," In Proceedings of Conference AIAP'2018: Artificial Intelligence and Its Applications, El-Oued, Algeria, 2018.
16. Benardos P.G. and Vosniakos G.-C., (2007), "Optimizing feedforward artificial neural network architecture," Engineering Applications of Artificial Intelligence 20, pp. 365382, 2007.
17. Benardos P. G. and Vosniakos G.C., (2002), "Prediction of surface roughness in CNC face milling using neural networks and Taguchi's design of experiments," Robotics and Computer Integrated Manufacturing 18, pp.343354, 2002.
18. Leski J. and Czogala E., (1999), "A new artificial network based fuzzy interference system with moving consequents in ifthen rules and selected applications," Fuzzy Sets and Systems 108, pp. 289297, 1999.
19. Jiang X. and Wah A.H.K.S., (2003), "Constructing and training feed-forward neural networks for pattern classification," Pattern Recognition 36, pp.853867, 2003.
20. Meyer D., Hornik K. and Feinerer I., (2008), "Text Mining Infrastructure in R," Journal of Statistical Software, 25 (5). pp. 154, 2008, in ISSN 1548-7660.
21. Agrawal R., Mehta M., Shafer J., Srikant R., Arning A. and Bollinger T., (1996), "The Quest Data Mining System," KDD-96 Proceedings, Systems for Mining Large Database, pp . 244-249, 1996.
22. Zaki M. J., Parthasarathy S., Ogihara M. and Li W., (1997), "Parallel Algorithms for Discovery of Association Rules," Data Mining and Knowledge Discovery, Volume 1, Issue 4, pp. 343373, 1997.

23. Brin S., Motwani R., Ullman J. D. and Tsur S., (1997), "Dynamic items and counting and implication rules for market basket data," *Proceedings of the 1997 ACM SIGMOD International Conference*, 1997, in 10.1145/253260.253325.
24. A. Ashir, (2012), "Use of Social Media in Disaster Management," In *Conference: Conference: ICITE 2012, At Hong Kong*, Volume: IPEDR, volume 39, 2012.
25. Hui, L. H. D. and Tsang, P. K. E. (2016), "Everyday Knowledge and Disaster Management: The Role of Social Media," *Everyday Knowledge, Education and Sustainable Futures*, Volume 30, pp. 107-121.
26. Chikoto G. L., Sadiq A. A. and Fordyce E., (2013), "Disaster Mitigation and Preparedness Comparison of Nonprofit, Public, and Private Organizations," *Nonprofit and Voluntary Sector Quarterly*, 42(2), pp. 391-410, 2013.
27. Estuar M. R. E., Ilagan J. O., Victorino J. N. N. C. Lagmay M. and Hechanova M. R. (2016), "The Challenge of Continuous User Participation in eBayanihan: Digitizing Humanitarian Action in A Nationwide Web Mobile Participatory Disaster Management System," In *3rd International Conference on 'Information and Communication Technologies for Disaster Management (ICT-DM)'*, Vienna, Austria.
28. Ogie R. I., Rho J. C. and Clarke R. J., (2018), "Artificial Intelligence in Disaster Risk Communication: A Systematic Literature Review," *5th International Conference on Information and Communication Technologies for Disaster Management (ICT-DM)*, 2018, in doi:10.1109/ict-dm.2018.8636380.
29. Fox E. A. and Shaw J. A., (1993), "Combination of Multiple Searches," *Proceedings of TREC 1993*. <http://trec.nist.gov/pubs/trec2/papers/txt/23.txt>, 1993.
30. Vulic I. and Moens M.-F., (2015), "Monolingual and Cross-Lingual Information Retrieval Models Based on (Bilingual) Word Embeddings," *SIGIR'15 Proceedings of the 38th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 363-372, Santiago, Chile — August 09 - 13, 2015.
31. Young S. D., Rivers C. and Lewis B., (2014), "Methods of using real-time social media technologies for detection and remote monitoring of HIV outcomes," In *Preventive Medicine*, Volume 63, p. 112-115.
32. Roy A., Ghosh K., Basu M., Gupta P. and Ghosh S., (2018), "Retrieving Information from Multiple Sources," *WWW'18*, April 23-27, 2018.
33. Islam, S. M. T. and Chik, Z. (2011), "Disaster in Bangladesh and management with advanced information system," In *Disaster Prevention and Management: An International Journal*, Vol. 20, Issue: 5, pp. 521-530, available in: <https://doi.org/10.1108/09653561111178952>.
34. Shaluf, I. M. (2008), "Technological disaster stages and management," *Disaster Prevention and Management: An International Journal*, Vol. 17, Issue 1, pp.114-126, IN: DOI:10.1108/09653560810855928.
35. Ross J.P., "Taguchi Techniques for Quality Engineering, (1996), " McGraw-Hill, New York, 1996.
36. Bharanipriya V. and Kamakshi Prasad V., (2011), "Web Content Mining Tools: A Comparative Study," *International Journal of Information Technology and Knowledge Management* January-June 2011, Volume 4, No. 1, pp. 211-215
37. Alam F., Joty S. and Imran M., (2018), "Graph Based Semi-supervised Learning with Convolutional Neural Networks to Classify Crisis Related Tweets," In *International AAAI Conference on Web and Social Media (ICWSM)*
38. Nazer T. H., Xue G., Ji J. Y. and Liu H., (2017), "Intelligent Disaster Response via Social Media Analysis - A Survey," *ACM SIGKDD Explorations Newsletter*, Vol. 19, No 1, p. 46-59